# From Declared Risk Levels to Admissible Recommendations: Measuring How Often an Intermediate Governance Level Binds in Deterministic Historical Replay

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***Abstract*— A declared risk level constrains an artificial intelligence (AI) system only if it changes what the system may recommend, yet its presence in a specification establishes neither that effect nor how often it binds. Existing approaches restrict system behaviour under changing conditions, but the reviewed literature does not characterise how often intermediate advisory-scope restrictions change admissible recommendations relative to participation-only gating. We define an operational criterion under which a declared level changes governance behaviour only if it changes the admissible recommendation set. To isolate the intermediate restriction, we compare configurations differing only at the intermediate state through deterministic historical replay of up to five years of hourly environmental records for a coastal-fisheries site in Kota Kinabalu, Sabah. Under a stipulated three-state classifier, the restriction bound in 5.81% of evaluated departure-window hours under PRIMARY and 4.48% under RESOLUTION. Binding frequency ranged from 5.81–9.22% across two source-grounded interpretations of the small-vessel wave UNSAFE boundary, fell to 1.26% under a large-vessel configuration, and differed by 1.97 percentage points between wave data products evaluated on identical hours. Historical marine-warning data were unavailable and held at the least-severe value; reported frequencies are conditional on this exclusion and are not necessarily lower bounds. These measurements characterise governance activation, not safety or decision outcomes, and demonstrate how declared governance levels can be audited through the recommendation constraints they impose and the conditions under which they bind.**

***Keywords*— *AI governance, decision support, admissible recommendations, coastal fisheries, historical replay***

## I. Introduction

Systems supporting consequential decisions are increasingly placed under explicit operating controls, commonly described by the risk levels they declare. A declared level constrains a system only if it changes the set of actions the system is permitted to take, and the presence of the level in a specification does not by itself establish that it does. Neither does it establish how often the constraint is encountered once the system runs against real conditions. These properties can therefore be treated as measurement questions rather than assumed from the design specification [1].

Mechanisms for restricting what a system may do are established: run-time assurance in aviation [2], pre-selection shielding in safe reinforcement learning [3], progressive advisory inhibition in certified collision-avoidance avionics [4], and the graduated contraction of alternatives described in the canonical model of human interaction with automation [5]. Section II treats these in more detail. This paper contributes none of these mechanisms.

That model also holds that automation cannot be assigned a single overall level, because a level is always stated relative to a specific function [5]. What separates one level from another is the allocation it changes, not its position in a taxonomy. Applied to a declared risk level, this yields a question that can be checked rather than assumed: does the level alter the set of recommendations the system may produce, or does it appear only in the specification? A level that leaves the permitted set unchanged describes the design without constraining it.

Measuring how often such a boundary is encountered is not an unexplored activity, and Section II identifies work that performs parts of it. The narrower case concerns an intermediate level that contracts the permitted set while the system continues to operate, examined against a configuration that gates participation alone. Within the reviewed literature, no empirical characterisation was identified of how often an intermediate advisory-scope level changes the admissible recommendation set relative to a participation-only gate when replayed over a multi-year environmental record. That statement is scoped to the literature examined here, and to the combination of properties just described, rather than being a claim about the field as a whole.

This study addresses one question. **RQ1.** Over a multi-year environmental record for one deployment site, how often is an intermediate advisory-scope governance level reached under a stipulated three-state classifier, and, the comparator difference being fixed by the admissible-set mappings, how sensitive is that frequency to defensible threshold, wave-data-product, vessel-class and evaluation-set configurations? The approach states a specification-level criterion and defines four configurations, $C_0$ to $C_3$, differing at one state only so the intermediate level can be isolated. They are replayed over up to five years of hourly records for a single coastal-fisheries site,

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and the frequency is examined across the defensible settings the evaluation permits.

The contribution is a measurement assembly and its interpretation rather than any of its constituent techniques. It includes the operational criterion for determining when a declared level changes governance behaviour; the interpretation of the resulting quantity as the frequency with which the active contraction binds, under the specified comparator construction; the sensitivity of that binding rate to the threshold, data product and evaluation set that govern it; the attribution of which state component produces the intermediate classification in the evaluated record; and two limits of the measure: it is invariant to the content of the restricted set, including the empty set, and it does not distinguish a three-state declaration leaving scope unchanged from a two-state gate. Section VI states what the evaluation does not establish.

## II. Background and Positioning

Reviews of AI governance literature report gaps in implementation detail and real-world validation, with few studies specifying together who governs, what is governed, when, and how [6]. Risk-management guidance calls for AI risks and impacts to be analysed, assessed, benchmarked and monitored, with testing before deployment and regularly during operation [1]. Operational measurement of deployed controls is therefore established rather than absent. The question here is narrower: whether one declared level has an observable effect on the recommendations a system may produce.

Run-time action-restriction mechanisms are well established. ASTM F3269 standardises run-time assurance bounding complex aircraft-system functions through an independently assured architecture [2]. Safe reinforcement learning distinguishes pre-selection shielding from post-selection correction [3]. Certified collision-avoidance avionics progressively inhibit advisory categories as conditions worsen while retaining crew authority [4]. These precedents establish that state-conditioned restriction is not the contribution of this study; what it examines is how the operational effect of a declared level can be measured.

Prior work has also quantified how frequently safety mechanisms intervene. For example, Corsi et al. [7]measured shield activation frequency during reinforcement-learning policy execution. In contrast, the present study characterises how often an intermediate environmental state imposes a stricter admissible recommendation set relative to participation-only gating, without evaluating generated recommendations.

The individuation of levels has a longer history in human-automation research. The canonical model holds that a level of automation is always stated relative to a specific function, and it describes graduated restriction of what is offered to a person across adjacent levels: a complete set of alternatives, a narrowed selection, or a single suggestion [5]. That model is design-time and defines no run-time state variable. It nonetheless supplies the grounding for examining a declared level through the behaviour it permits or suppresses, rather than through its position in a scheme, which is the criterion stated in Section III.

Three recent studies each perform part of the measurement assembled here. Aarab specifies graduated run-time autonomy modes in which the permitted execution directives contract as severity rises while operation continues [8]; the reported directive-frequency simulation is conducted under the permissive mode rather than as a comparison of graduated configurations. Miller examines governance enforcement through approval-only strategies and projection-based repair [9]; this comparison concerns enforcement approaches rather than graduated risk levels and does not establish how often an intermediate advisory-scope contraction binds. Kim et al. quantify the burden of an intermediate zone requiring mandatory human review, and characterise it across reliability targets on real clinical data [10]; that zone withdraws automated action rather than retaining an active contracted recommendation set. Within the reviewed literature, these works do not combine active intermediate scope contraction, a participation-only comparator, marginal isolation, and binding-frequency measurement.

## III. Governance Contract, Criterion and Metric

Fig. 1 summarises the evaluation framework: records are classified into a three-state variable, a governance contract maps each state to an admissible recommendation set, four configurations instantiate that contract with and without the intermediate contraction, and the configurations are replayed over the same classified record. This section states the governed object, the criterion and the metric formally; Section IV states the data, the provenance of each boundary and the replay procedure.

Evaluating the operational effect of an intermediate governance restriction requires the governed object to be stated explicitly. We take it to be $A_{AI}(S)$, the set of recommendation types admissible when the classified operating state is $S$. Each evaluated hour is assigned one of three states, SAFE, CAUTION or UNSAFE, and FULL denotes the unrestricted recommendation set {Go, Delay, DepartureTime, Duration}. Table I states the contract. The admissible sets contract monotonically across the three states by construction. AI participation is represented by the state-dependent gate $G(S)$, while $A_{AI}(S)$ specifies the admissible recommendation set. Under the evaluated contract, $G(S) = 0$ and $A_{AI}(S) = \emptyset$ at UNSAFE; at CAUTION, $G(S) = 1$ and $A_{AI}(S)$ is a non-empty strict subset of FULL.

$A_{AI}$(CAUTION) = {Go, Delay} is a researcher-defined design choice used to instantiate a strict intermediate contraction for the evaluation. A strict subset of FULL is required at the intermediate state because a level that leaves the admissible set unchanged instantiates no contraction, and so provides nothing to isolate or measure. As the consequences of the identity below make explicit, the measurement reported in this paper does not validate it.

TABLE I. Admissible recommendation set by safety state.

| State | $G$ | $A_{AI}(S)$ |
|---|---|---|
| SAFE | 1 | FULL |
| CAUTION | 1 | {Go, Delay} |
| UNSAFE | 0 | ∅ |

$G$ denotes the state-dependent AI participation gate (1 = enabled; 0 = disabled). The evaluated governance configuration satisfies $\emptyset \subsetneq A_{AI}(\text{CAUTION}) \subsetneq \text{FULL}$. The CAUTION admissible set, $A_{AI}$(CAUTION) = {Go, Delay} is researcher-defined; it has not been empirically validated or demonstrated to be optimal or safer.

Following the functional interpretation of levels set out above, we evaluate a declared state through the recommendation set it permits.

**Criterion.** A declared risk level changes governance behaviour only if it changes the admissible recommendation set. Two configurations that assign identical admissible sets at every state denote the same governance function, irrespective of how many states each declares.

This is an operational measurement rule rather than a result. It restates, for a declared state, the position already established in the levels-of-automation literature that a level is meaningful through the function allocation it changes [5].

Four configurations are defined over the same classifier, and their admissible-set mappings are given in Stage 4 of Fig. 1. $C_0$ applies no restriction. $C_1$ gates participation alone. $C_2$ is the evaluated intermediate-scope configuration. $C_3$ declares three states without contracting scope at the intermediate one.

$C_1$ and $C_2$ are identical at SAFE and at UNSAFE and differ only at CAUTION, where $C_1$ admits FULL and $C_2$ admits a strict subset. Any difference therefore arises exactly where the intermediate state contracts the admissible set, which supplies the isolation structure used below.

A three-state configuration that does not contract scope at the intermediate state, such as a traffic-light topology in which the intermediate level raises supervisory intensity rather than restricting what the AI may recommend [11], assigns the same admissible sets at every state as a two-state participation gate. Under the criterion the two are the same governance function, and the criterion therefore cannot distinguish them: a governance difference expressed as supervisory intensity rather than admissible scope lies outside what this metric observes. Declared state cardinality is not by itself evidence of a governance effect.

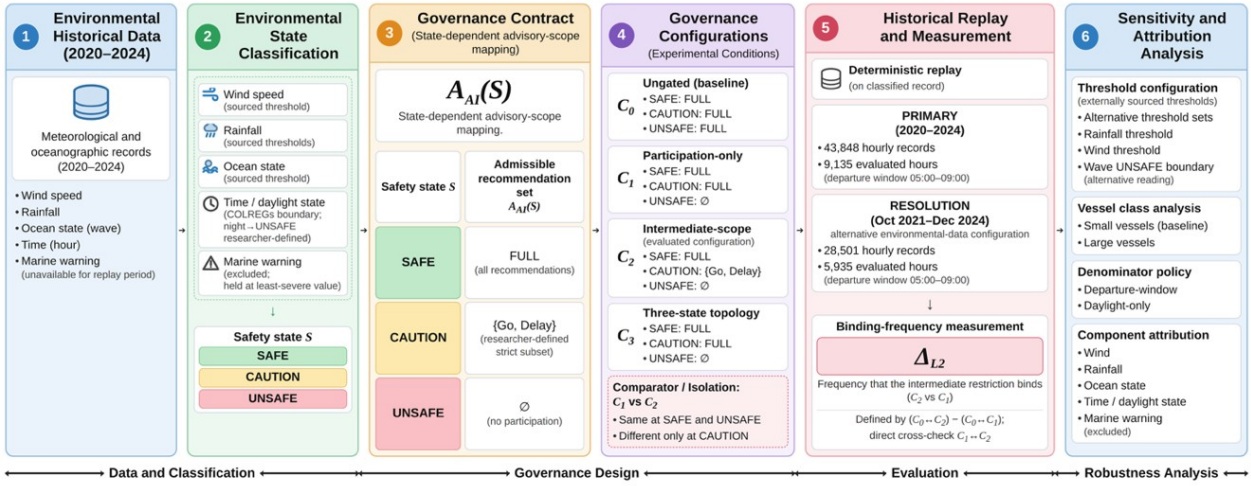



Fig. 1. Evaluation framework for measuring how often the intermediate governance restriction binds.

Let $N_{\text{eval}}$ be the number of evaluated departure-window hours and $N_{\text{restricted}}$ the number of those hours in which one configuration admits a strictly smaller set than another. The admissible sets compared in this study are nested, so each observed difference represents a strict restriction in one direction. The difference rate between two configurations is

$$\Delta = (N_{restricted} / N_{eval}) \times 100\%.$$

The isolated quantity is

$$\Delta_{L2} = (C_0 \leftrightarrow C_2) - (C_0 \leftrightarrow C_1).$$

Here $C_0 \leftrightarrow C_1$ captures participation gating alone and $C_0 \leftrightarrow C_2$ captures both levels, so their difference isolates the increment associated with the intermediate level. A direct $C_1 \leftrightarrow C_2$ comparison is computed as a cross-check.

The identity governing interpretation follows from the mappings rather than from the data. $C_1$ and $C_2$ agree at SAFE and at UNSAFE and differ at CAUTION, where $C_2$ admits a strict subset of FULL. The two configurations therefore differ on exactly the CAUTION hours, so that

$$\Delta_{L2} \equiv P(S = CAUTION)$$

for the evaluated mappings. This is a metric-interpretation identity, not a result.

Two consequences follow. One is that the value of $\Delta_{L2}$ is invariant to the content of $A_{\text{AI}}$(CAUTION), provided that set remains a proper subset of FULL, including the empty set: replacing one such subset with another leaves the binding frequency unchanged when the state classification is unchanged, and non-emptiness is a property of the declared contract rather than something the metric detects. The other is that $\Delta_{L2}$ measures how often the intermediate restriction binds, not how much the recommendation set contracts.

Because the identity holds, the CAUTION base rate yields the same binding frequency, and the comparator ladder is not required to compute it. The ladder instead separates participation gating from intermediate scope contraction and verifies that the isolated difference occurs only at CAUTION. The empirical contribution is therefore not the identity itself, but the characterisation of how frequently this restriction binds in the historical record, which environmental components produce the intermediate classification, and how the measured frequency changes across the evaluated configurations.

## IV. Evaluation Design

The replay covers one deployment site, Kota Kinabalu, Sabah, a low-resource coastal-fisheries setting, with a small-vessel baseline under 10 gross register tonnage (GRT). The wave CAUTION onset is informed by a length-based restriction for vessels under 10 m [17]; its application to the tonnage-defined baseline is a researcher-defined transfer of a length-parameterised guideline rather than an established equivalence between vessel classes.

Two environmental-data configurations are evaluated. The PRIMARY configuration covers January 2020 to December 2024, comprising 43,848 hourly records with wave data from the European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis, Fifth Generation (ERA5), at approximately 50 km resolution, giving a departure-window denominator of 9,135 hours. The RESOLUTION configuration covers October 2021 to December 2024, comprising 28,501 usable hourly records and 5,935 departure-window hours. Eleven initial wave observations were unavailable and excluded. Wave data came from the Open-Meteo Marine API with the meteofrance_wave model, documented as Météo-France Wave Model (MFWAM) at about 8 km resolution [12]. Records were retrieved for 5.940 N, 116.025 E (weather), 6.00 N, 116.00 E (ERA5-Ocean waves) and 5.958 N, 116.042 E (MFWAM waves), with sea cell selection and timezone Asia/Kuching. Weather records were retrieved through the Open-Meteo archive [13], [14].

The evaluation window is the five hourly timestamps from 05:00 to 09:00 inclusive local time (UTC+8), which defines $N_{\text{eval}}$. The window represents a researcher-defined early-morning departure scenario rather than an empirically observed departure period, and other denominators are defensible.

Each hour is assigned a state using the maximum severity among the available component classifications, covering wind, rainfall, marine warning, ocean state and time. Vessel class parameterises the ocean-state component rather than contributing an independent severity term.

Threshold provenance is separated from design choice. The wind boundary follows Malaysian Meteorological Department strong-wind criteria [15], and the rainfall UNSAFE boundary follows the same department's thunderstorm criteria [16]. For the small-vessel class, the ocean-state CAUTION onset is 1.0 m, following the departure restrictions or cautions proposed for vessels under 10 m in length in [17], of which the non-prohibitive reading is adopted; the UNSAFE boundary is 1.25 m, the upper limit of the highest sea-state category passed by the 6.54 m hull tested in [18]. Reference [17] sets no upper boundary and neither source is keyed to tonnage, so the CAUTION band is a stipulated parameter composed from two source-supported edges rather than a validated transition. The daylight boundary uses the sunset-to-sunrise interval specified in International Regulations for Preventing Collisions at Sea (COLREGs), Rule 20(b) [19]; treating non-daylight hours as UNSAFE is a researcher-defined governance policy, not a departure prohibition established by that rule. The rainfall CAUTION boundary is source-supported through the Department of Irrigation and Drainage Malaysia (DID) light-rainfall limit [20] but is not a Meteorological Department threshold, since no criterion is published below the thunderstorm level. Source provenance indicates where a boundary comes from; it does not establish that it is optimal. Researcher-defined choices are the three-state cardinality, $A_{\text{AI}}$(CAUTION), the policy treating non-daylight hours as UNSAFE, the evaluation window, and the declared exclusion described next.

The component boundaries are: wind CAUTION above 21.6 kn, UNSAFE above 27.0 kn; rainfall CAUTION above 10.0 mm/h, UNSAFE above 20.0 mm/h or on World Meteorological Organization (WMO) weather codes 95, 96 and 99; small-vessel ocean state CAUTION at or above 1.0 m, UNSAFE above 1.25 m; time UNSAFE outside the interval from sunrise to sunset; and marine warning held at SAFE. Comparisons are strict except the ocean-state CAUTION onset, which is inclusive, following the cited phrasing “1 m or above” [17]. The classified state is the maximum over the five components under SAFE < CAUTION < UNSAFE, and a missing clock, date or solar lookup resolves to UNSAFE. Solar events come from a frozen daily table for 5.98 N, 116.01 E at UTC+8, exact sunrise SAFE and exact sunset UNSAFE. These conventions are stated for exact reproducibility, not as claims that the boundaries are validated for the baseline class.

The two kinds of choice act at different points: environmental thresholds determine when a state such as CAUTION occurs, while the governance contract determines what may be recommended once it holds. $A_{\text{AI}}$(CAUTION) is a choice of the second kind, and the evaluation measures how often the contraction it defines binds rather than whether its content is appropriate.

The marine-warning component has no usable historical archive for the replay period. It is handled by declared exclusion rather than imputation, and held at its least-severe value throughout. Because the classifier aggregates by maximum severity, an unavailable marine-warning observation could maintain or increase the classified severity. Consequently, CAUTION-or-higher activation cannot decrease when the excluded component is restored. However, CAUTION occupancy alone may increase or decrease because additional warnings can move hours into or out of CAUTION. The reported $\Delta_{L2}$ values are therefore conditional on the declared exclusion and should not be interpreted as lower bounds.

Classification remains total under the exclusion: every evaluated hour receives a state from the available component rules and the declared treatment of the excluded component.

The procedure is a deterministic historical replay rather than a model-training experiment. For each configuration, every evaluated hour in the specified record is classified by the component rules, the configuration's admissible-set mapping is applied to the resulting state, and the hours on which two configurations admit different sets are counted. $\Delta_{L2}$ follows from those counts by the definitions in Section III. The same operations are repeated for each alternative threshold, vessel-class and denominator setting reported in Section V. Because the replay is deterministic, no configuration needs replication. The reported figures are exactly repeatable from the archived input records, which are not guaranteed to be re-retrievable from the third-party interface that serves them. The canonical sensitivity script performs 78 evaluations covering 56 distinct combinations of environmental-data configuration, threshold setting, and evaluation-set policy (departure window, daylight-only denominator, or common-period restriction). The four governance configurations are applied to each classified record rather than replayed separately. No parameter is estimated from the data and no randomisation is involved, so replaying an identical record under an identical configuration reproduces an identical result.

The replay is a deterministic census of all evaluated departure-window hours in the specified records. No sampling inference is performed, and no p-values, confidence intervals or significance claims apply. Ranges reported later are deterministic configuration ranges rather than confidence intervals.

## V. Results and Discussion

### A. Binding Frequency and Comparator Decomposition

The PRIMARY replay gave $\Delta_{L2}$ = 5.81% and the RESOLUTION configuration gave 4.48%. These are proportions of evaluated departure-window hours in which the intermediate advisory-scope restriction binds. As established in Section III, this equals CAUTION occupancy under these mappings.

The PRIMARY comparator decomposition separates the two levels. $C_0 \leftrightarrow C_1$ gave 42.88%, the frequency associated with participation gating alone, and $C_0 \leftrightarrow C_2$ gave 48.69%, the frequency associated with participation gating together with the intermediate contraction. The difference, 5.81 percentage points, isolates the incremental binding frequency. The direct $C_1 \leftrightarrow C_2$ comparison returned the same value. The RESOLUTION decomposition gives $C_0 \leftrightarrow C_1$ = 41.08% and $C_0 \leftrightarrow C2$ = 45.56%, a difference of 266 of the 5,935 evaluated hours. The participation-only disagreement is dominated by the researcher-defined non-daylight policy: 86.5% of PRIMARY UNSAFE hours and 93.2% of RESOLUTION UNSAFE hours were classified UNSAFE solely because they occurred before sunrise, without another UNSAFE trigger.

The two configurations differ in both wave source and record period, and the gap between 5.81% and 4.48% indicates that the measured binding frequency depends on the environmental-data configuration used for replay. Neither value is treated here as more accurate than the other, and the difference is not attributed to spatial resolution alone. Evaluated on the 5,935 hours common to both records, with identical thresholds and denominator, the two products give 6.45% and 4.48%. The full-record gap is 1.33 percentage points, compared with 1.97 points on the common period, reflecting a 0.64-point difference between the full PRIMARY record and its common-period subset.

TABLE II. Binding frequency, comparator decomposition and sensitivity results

| Metric | PRIMARY | RESOLUTION |
|---|---|---|
| Departure-window hours | 9,135 | 5,935 |
| $C_0 \leftrightarrow C_1$ | 42.88% | 41.08% |
| $C_0 \leftrightarrow C_2$ | 48.69% | 45.56% |
| $\Delta_{L2}$ | 5.81% | 4.48% |
| Wave-boundary interpretation range | 5.81–9.22% | 4.48–6.08% |
| Small-vessel baseline | <10 GRT | <10 GRT |

The wave-boundary interpretation range is a deterministic configuration range, not a confidence interval.

### B. Threshold Sensitivity

Across two source-grounded interpretations of the small-vessel wave UNSAFE boundary, with rainfall and wind thresholds held at their canonical settings, $\Delta_{L2}$ ranges from 5.81% to 9.22% in PRIMARY and from 4.48% to 6.08% in RESOLUTION. The measured binding frequency is therefore sensitive to the operationalisation of the wave boundary.

Provenance does not by itself remove designer interpretation. Two readings of the small-vessel wave UNSAFE boundary were evaluated: the upper limit of the highest passing sea-state category (1.25 m) and the tested failing condition (1.875 m, implemented as 1.90 m), giving $\Delta_{L2}$ of 5.81% and 9.22%. Both derive from [18], and neither is established as optimal.

Varying the wave UNSAFE boundary across its two source-grounded readings moves $\Delta_{L2}$ by 3.41 percentage points in PRIMARY, while varying the rainfall CAUTION boundary across the exploratory range evaluated here changes $\Delta_{L2}$ by 0.48 percentage points in PRIMARY and 0.78 percentage points in RESOLUTION. The wave boundary was varied only over sourced values and the rainfall boundary over an unsourced exploratory range, so the two movements are not comparable as a ratio and establish no intrinsic sensitivity ranking beyond the ranges evaluated. At the sourced wind settings the component binds in no evaluated hour, the maximum departure-window wind being 21.80 kn in PRIMARY and 19.30 kn in RESOLUTION against an UNSAFE boundary of 27.0 kn; varying the wind CAUTION boundary across the evaluated range $W_C$ = 18–25 kn produced no change in $\Delta_{L2}$ in either environmental-data configuration.

### C. Component Attribution and Scope Dependence

Within the PRIMARY departure window, 531 hours were classified CAUTION, and attribution across the component classifiers is uneven. The ocean-state component accounted for 518 of those 531 hours, or 97.6%, and the rainfall component for the remaining 13, or 2.4%. These are attributions of CAUTION classification, not of incidents or unsafe outcomes, none of which is observed here. Because the ocean-state component produces almost all of these hours, the reported frequency rests substantially on the stipulated wave band and on the length-to-tonnage transfer declared in Section IV.

The three remaining components register zero, for three different reasons, and the distinction matters. No PRIMARY CAUTION hour was attributed to wind, which is an empirical negative finding for the evaluated record and is separate from the structural zero reported for the wind threshold range above. The time component emits no CAUTION by construction, so its zero is definitional. The marine-warning component is structurally excluded, and its zero reflects unavailability rather than evidence that marine warnings never bind.

The measured frequency also depends on configuration choices fixed in Section IV. Under the large-vessel threshold configuration the PRIMARY value falls to 1.26%, and to 0.37% in RESOLUTION. Restricting the evaluated departure-window hours to daylight gives denominators of 5,536 and 3,594 hours. Because a CAUTION classification requires that the time component not be UNSAFE, every CAUTION hour already lies inside the daylight subset, so the numerator is unchanged at 531 and 266 and the rate rises to 9.59% and 7.40% by denominator reduction alone. Vessel class and evaluation set are policy settings; neither variant is presented as more appropriate than the baseline. Because the marine-warning component was held at its least-severe value, the reported binding frequencies are conditional on its exclusion and are not necessarily lower bounds.

RQ1 can be answered conditionally. For the PRIMARY configuration at the evaluated site and small-vessel baseline, the intermediate advisory-scope contraction binds in 5.81% of departure-window hours, and in 4.48% under the RESOLUTION configuration. That frequency is conditional rather than fixed: it spans 5.81–9.22% across two source-grounded wave-boundary interpretations, falls to 1.26% for the large-vessel class, and rises to 9.59% under a daylight-only denominator. The intermediate level therefore measurably changes the admissible recommendation set under the evaluated contract, with the frequency of that change depending on

threshold, wave-data-product, vessel-class and evaluation-set choices, under the stipulated classifier whose intermediate band is not validated for the baseline vessel class.

The rate cannot be read without also examining threshold provenance, component attribution, vessel-class assumptions, evaluation-set policy and the excluded component.

## VI. Limitations and Conclusion

The replay concerns one deployment site and should not be read as a population estimate for other locations. The activation frequency is conditional on threshold interpretation, with the two source-grounded interpretations of the small-vessel wave UNSAFE boundary producing a larger change in binding frequency than the evaluated rainfall and wind threshold variations. Several governance choices remain researcher-defined rather than empirically validated: the three-state cardinality, the evaluation policies fixed in Section IV, the intermediate ocean-state band, which is composed from two sources keyed to vessel length rather than tonnage and is not validated for the baseline class, and the admissible set $A_{\mathrm{AI}}(\mathrm{CAUTION}) = \{\mathrm{Go}, \mathrm{Delay}\}$. The evaluation measures how often that restriction binds. It does not establish that {Go, Delay} is the correct, optimal or safer set at CAUTION, and the reported measure is invariant to its content. The marine-warning component has no usable archive for the replay period and was held at its least-severe value. The reported binding frequencies are therefore conditional on this exclusion and are not necessarily lower bounds.

The study characterises governance activation; it does not validate it. No incident or outcome record exists for the site, so activation is not linked to unsafe departures or avoided harm. No generated advice is evaluated: the metric ranges over admissible recommendation types, not over the correctness, quality or usefulness of any recommendation. Future work could evaluate counterfactual binding by testing how often generated recommendations would fall outside the admissible set under CAUTION, rather than measuring only when the restriction is active. No user study has been conducted, so interpretation, acceptance, trust, behavioural response and human oversight in use remain untested. The replay is a deterministic census rather than a sample, so no statistical inference applies, and the vessel-class variant is a configuration check, not evidence of generalisation across classes, sites or periods. Before any claim of safety or generalisation is made, field work should test the threshold mappings and the restricted set with domain stakeholders, examine how users interpret the advice, and relate activation to operational outcomes.

The question addressed was whether a declared intermediate risk level changes what an advisory system is permitted to recommend, and how often. Stating a criterion on admissible sets and isolating the intermediate level with a comparator that differs at one state, the contracting level was reached in 5.81% of PRIMARY departure-window hours and 4.48% under RESOLUTION. These frequencies are conditional on the evaluated threshold, vessel-class and data-product configurations rather than stable site properties. The replay establishes an activation frequency for one contract at one site under one classifier; by construction the isolated increment equals the intermediate state's occupancy, so the observed binding frequency depends on the classifier, environmental record and evaluation policy, while the admissible-set contract determines what that frequency represents; and the wave band, the admissible set at CAUTION, and any link between activation and outcome remain unvalidated. The contribution is the assembly that makes such a figure interpretable, together with the criterion. For Responsible AI governance, the implication is that a declared risk level can be examined not only by its label, but by the admissible recommendations it changes and the conditions under which that change binds.

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